Telecoms • End-to-end RAN + Core • 2G/3G/4G/5G • IMS • Transport • Cloud • OSS/BSS

Frutos Kamanzi

Senior End-to-End RAN & Core Network Engineer

Independent consultant with 18+ years end to end, radio to core: BTS/BSC, NodeB/RNC, eNodeB, gNodeB and EN-DC; CS core, EPC, 5GC SBA, IMS/VoLTE; SS7/SIGTRAN, Diameter, SIP, HLR/UDM; cloud native and optimisation. MTN, Tigo, Airtel and Ericsson Managed Services, with platform depth in ZTE and Ericsson.

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RAN TO CORE, END TO END • REAL PROBLEMS SOLVED • RESULTS DELIVERED

FELICIA
INTELLIGENT TELCO ENGINEERING

AI-POWERED PORTFOLIO

From Rwanda to the world



18+ Years
Telecom experience · multi-vendor, multi-operator

25,000+ Words
of public diagnosis · 135 answers, 56 threads

69 Follow-ups
exchanges with practising engineers

11 Badges
community awards · incl. New User of the Month

3 Corrections
made in public, left on the record

14 Full cases
in the casebook · + 15 engagements, 5 advisory

Career, 2007 to today

2007 – 2013
Switch engineer, core and RF
MTN Rwanda · Tigo Rwanda

2013 – 2015
Standards & international roaming engineer
Ericsson Rwanda

2015 – 2023
Core network engineer & acting operations manager
Airtel Rwanda

2023 – 2025
CS core SME lead & acting operations manager
Ericsson MS · MTN Rwanda

2025 – 2026
Senior core network expert (L6)
ZTE core · Airtel Rwanda

2026+
Independent consultant and remote expert
Kigali · worldwide

99.95% CS core network availability
sustained, 1+1

–35% recurring P2/P3 incidents
after process redesign

4 h → 15 min
diagnosis and RCA time, with automation I built

End to end: RAN, transport, core, IMS. Vendor neutral.

End to end, radio to core

One fault domain at a time, across the whole chain. Every block is backed by public cases or production work.

Device / UE
Handsets, modems, eSIM and eUICC, JavaCard profiles
Proof: 37073 handset test · eUICC/JavaCard answers

RAN
2G BTS/BSC · 3G NodeB/RNC · 4G eNodeB · 5G gNodeB · EN-DC · BBU/RRU · mobility, interference, licensing
Proof: 15161 · 36955 · 36886 · 36820 · 36924

Transport
S1 / NG and S1-U / N3 user plane, SCTP, IP backhaul, early VSAT international links
Proof: 37013 S1-U endpoint

Core
CS: MSC-S, MGW, HLR · EPC: MME, SGW, PGW · 5GC SBA: AMF, SMF, UPF, UDM, PCF, NRF
Proof: 54-node CS core · 5G SA lab · Open5GS #4672 #4699

IMS and voice
P/S/I-CSCF, TAS, VoLTE, SRVCC, IMS authentication
Proof: 36843 SRVCC · VoLTE rollout

Signalling and roaming
SS7/SIGTRAN, MAP, Diameter, SBI HTTP/2, SMS and OTP, steering of roaming
Proof: 36930 · Osmocom #7048 · SIGTRAN MOP

RAN vendor depth

ERICSSON SRVCC B2, EN-DC and S1-U cases in public · RAN–core boundary of a 54-node CS core: BSC, RNC, eNodeB, gNodeB

ZTE eNodeB traffic into a ZTE packet core over S1-MME and S1-U, in production

ANY VENDOR Huawei RAN worked from Huawei documentation: BBU5900, LTE handover, S1 cause 29

Plus the site underneath: grid power, generators, batteries and solar, with energy baselines for 1,467 shared sites.

CS Core estate, Ericsson and ZTE

Ericsson Managed Services, MTN Rwanda account · CS core SME lead · 2023 – 2025 · then ZTE iCX and iMG on TECS · 2025 – 2026

54 Ericsson nodes
CS core · 1+1 on all critical nodes
MSC-S | MGW | HLR | STP | SG

665,000 Erlang
traffic capacity handled
15 min P1 SLA · live-change governance

CS core sites
GAC primary · REM secondary · HUY regional, with 1+1 redundancy on all critical nodes.

Intelligent telecom engineering
Felicia labels every statement by its source.
- ✓ Document evidence
- ✓ Live-lab evidence
- ✓ Engineering inference
- ✓ Not proven, stated

RAN
2G | 3G | 4G | 5G
BSC, RNC, eNodeB, gNodeB at the RAN–core boundary

Core network, Ericsson
MSC-S | MGW | HLR | VLR | STP | SG | GMSC

IP / SIGTRAN
M3UA · SCTP · MAP

External
PSTN | PLMN · roaming | interconnect

The method, applied in public
Scope › Evidence › Cause › Fix › Verify

- ✓ SIGTRAN routing MOP, ABI Chennai/Mumbai · no incident
- ✓ VLR dual-active self-healing MOP (GACMSC1/REMMSC1)
- ✓ Monthly KPI governance reports and MOPs
- ✓ SS7/SIGTRAN, MAP, Diameter, GTP, SIP, IMS

CS core vendor depth

ERICSSON 54 nodes: MSC-S, MGW, HLR, STP, SG · 665,000 Erlang · SIGTRAN and VLR MOPs · 2023 – 2025

ZTE iCX MSC server V5.17.20 and iMG media gateway V5.17.19, cloud-native on TECS · 2025 – 2026

ANY VENDOR Osmocom SS7/SIGTRAN defect reported (#7048) · Huawei and Nokia libraries through the same ID gate

PS Core & 5GC estate, ZTE and Ericsson

Senior core network expert (L6) · Airtel Rwanda network · 2025 – 2026 · plus a live 5G SA reference lab

xGW packet gateway

SGW · PGW · GGSN · UPF roles

V7.25.40, in production

S5/S8 · Gn/Gp · N3 · N6

uMAC mobility

MME · SGSN · V7.25.40

Attach, TAU/RAU, paging and bearer control

S1-MME · S11 · S6a

DSC Diameter core

Diameter routing · V7.25.40

Policy, charging and subscriber signalling

Gx · Gy · S6a · Rx

Cloud-native platform

TECS on OpenStack and Kubernetes: gateways and signalling run as virtualised and containerised network functions.

- ✓ CNF lifecycle
- ✓ CaaS / IaaS
- ✓ xGW, uMAC, DSC counters verified
- ✓ OpenStack · Kubernetes

RAN

eNodeB | gNodeB | EN-DC

S1 / NG control and user plane into the packet core

Control plane

MME · SGSN · AMF · SMF · PGW-C · SGW-C

User plane

SGW-U · PGW-U · UPF · N4 / PFCP

Policy & charging

PCRF · PCF · OCS · CHF · Gx / N7 · Gy / N40

Proven in public and in the lab

37073: RAU, SGSN context pull and PDP modify walked end to end.
37013: S1-U endpoint traced to CUP_Inner, F-TEID on UDP 2152.
Open5GS #4699 UPF PFCP crash · #4672 SMF QoS-flow crash.
5G SA lab: UPF failure injected, recovery in about 12 s.

- ✓ GTPv1/v2, PFCP/N4 and NAS cause-value decoding · 17 protocol families
- ✓ PDP context, EPS bearer, PDU session and QoS-flow RCA
- ✓ Session, attach and bearer KPIs traced to verified counters
- ✓ Policy and charging paths: Gx, Gy, N7, N40

PS core vendor depth

ERICSSON Packet cases in public: 4G-to-2G data loss after RAU and PDP modify (37073), S1-U user-plane endpoint (37013)

ZTE xGW gateway, uMAC mobility and DSC Diameter in production on TECS, OpenStack and Kubernetes · 2025 – 2026

ANY VENDOR Open5GS 5G SA core live: AMF, SMF, UPF · Huawei and Nokia libraries through the same ID gate

Subscriber Data Management, HLR · HSS · UDM

HLR, HSS, UDM, AUSF and EIR across CS, EPC, IMS and 5GC · Ericsson and ZTE production estates, open-source cores and the 5G SA lab

HLR

Ericsson HLR in production

54-node CS core, 1+1 on critical nodes; MAP over SIGTRAN

MAP · C / D · SIGTRAN

HSS

EPC and IMS subscriber data

Location, auth and service profiles; on ZTE, Diameter reaches it through the DSC routing agent in production

S6a · S6d · Cx · Sh

UDM / AUSF

5G-AKA · EAP-AKA' · SUCI

Live on Open5GS; AKMA built on free5GC across AUSF and UDM

N8 · N12 · N13

EIR

IMEI check · S13 · N5g-eir

Diameter S13 equipment check designed for the Open5GS MME; MAP Check IMEI on F

S13 · N5g-eir · MAP F

Data layer

UDR · FE / BE split

UDM front end over a UDR back end, with NRF and SCP discovery, in the 5G SA lab

Protocols

MAP · Diameter · SBI HTTP/2 · SIGTRAN · SCTP

Auth

5G-AKA · EAP-AKA' · SUCI · AKMA

Roaming data

SoR · ODB · VLR / MME registration

Proven in public and in the lab

36930: roaming SMS/OTP loss placed on HLR/HSS subscriber state: ODB, stale VLR/MME registration, SoR.
AKMA, 3GPP Rel-17: AAnF, AUSF and UDM on free5GC, TS 33.535 vectors, paper co-authored.
Open5GS: AMF–AUSF auth-vector decode hardening · S13 EIR design.
5G SA lab: UDM, AUSF, UDR and NRF live, 6/6 baseline.

- ✓ Subscriber state in HLR, HSS and UDM: barring, registration, roaming steering
- ✓ Authentication vectors and key hierarchy: 5G-AKA, EAP-AKA', AKMA
- ✓ Diameter S6a, S13 and Cx via DSC routing; MAP via STP
- ✓ UDM, AUSF, UDR, NRF and SCP live in the 5G SA lab

SDM vendor depth

ERICSSON HLR in production in the 54-node CS core, MAP over SIGTRAN via STP · 2023 – 2025

ZTE DSC Diameter routing agent in production, the path to the HSS · 2025 – 2026

ANY VENDOR Open5GS UDM, AUSF, UDR live · AKMA on free5GC: AAnF, AUSF, UDM · S13 EIR design

Manual-to-evidence pipeline, for any vendor's library

Library in, verified evidence, deliverables out. v1 2023 – 2025: 18,131 counters → v2 2026: 72,430 verified counters + 1,255 alarm codes.

1 Vendor library
- ZEDX, ALEX / CPI, HedEx
- PDF, HTML, DOCX, XLSX, CSV

2 Adapter
- Unpacks the archive format
- Walks the topic tree

3 Topic parser
- Counters, alarms, commands
- Per node and release

4 ID gate
- Vendor ID grammar, exact
- ZTE: C+9 / P+9 digits

5 Verified dictionary
- SQLite FTS5 + semantic search
- Nothing invented

6 Evidence out
- RCA, MOP + rollback
- Evidence packs, KPI correlation

End-to-end run · vendor library → verified evidence → deliverables

```
PS C:\Felicia> python zedx_extractor.py --discover
  [discover] vendor library root found · .zedx archives · DITA topic tree
  [discover] NE types: uMAC · xGW · DSC · iCX · iMG · CIIA · TECS
PS C:\Felicia> python zedx_extractor.py --types counters,alarms --out-dir C:\Felicia\extracts
  [unpack]   Performance Counter Reference · Alarm Handling · per node and release
  [gate]     ID grammar exact: C + 9 digits · P + 9 digits (CIIA/TECS) · rejects → rejects.csv
  [verify]   independent re-validation of every record ..................... PASS
  ✓ counters_extract.csv   72,430 verified counter IDs
  ✓ alarms_extract.csv      1,255 alarm codes
  [index]    watchdog → felicia.db · SQLite FTS5 + semantic embeddings · hybrid retrieval
felicia@lab:~$ ftl start
  ✓ 5G SA lab up · AMF SMF UPF AUSF UDM UDR PCF NSSF NRF SCP · gNB/UE · port 8787
felicia@lab:~$ felicia ask "UPF restart: sessions drop. How fast do they recover?"
  DOCUMENT EVIDENCE      xGW/UPF counters resolved to manual topics, C+9 verified
  LIVE LAB EVIDENCE      UPF fault injected · UE_UPF + UE_Internet fail · recovery ~12 s
  ENGINEERING INFERENCE  first deviation → cause value → counterfactual, marked as reasoning
  NOT PROVEN             DNS check is container-level only: stated, never assumed
PS C:\Felicia> zctl ingest --cmdlog --alarms --kpi .\case\UPF_Restart
  ✓ CMDLOG, alarm log and KPI export timestamp-aligned into the evidence store
PS C:\Felicia> zctl deliver --case UPF_Restart --rca --evidence-pack --mop
  ✓ RCA_UPF_Restart_v1.0_30Sep2026.docx          10 sections · causal chain · fix · rollback
  ✓ EvidencePack_UPF_Restart_v1.0_30Sep2026.xlsx  alarms · counters · KPI · P1/P2 · sources
  ✓ MOP_UPF_Restart_v1.0_30Sep2026.md            pre-checks · steps · rollback · T+30m/24h/7d
  Prepared by Frutos · nothing invented · every claim traceable to manual, trace or lab
PS C:\Felicia>
```

Deliverables out

DOCX RCA report
10 sections: evidence, alarms, CMDLOG, manual refs, causal chain, fix, rollback, verification, closure

XLSX Evidence pack
Alarms · counters · KPI trend · P1/P2 ranking · source authority · timestamp alignment

MD MOP with rollback
Pre-checks · ordered steps · rollback triggers · checks at T+30 min, 24 h, 7 d

JSON Case file
Every claim tied to a manual page, a trace line or a lab run

Same run for any vendor library: only the adapter and ID grammar change.

Vendor support and expertise

One verified gate for every vendor library: only the adapter and the ID grammar change

ZTE ZTE
iCX MSCS V5.17.20 · iMG MGW V5.17.19
xGW · uMAC · DSC V7.25.40 · CIIA V8.25.40 · TECS
Proven · 72,430 + 1,255

ERICSSON Ericsson
RAN | Core | IMS | Cloud
2G | 3G | 4G | 5G · ALEX / CPI libraries
Tested · gate held

HUAWEI Huawei
HedEx documentation packages
Huawei ID grammar at the gate
Adapter ready

NOKIA Nokia
Product documentation sets
Nokia ID grammar at the gate
Adapter ready

ANY Any vendor
PDF · HTML · DOCX · XLSX · CSV
multi-vendor, migration, interop
Built

Public casebook, July to September 2026

Real issues. Real fixes. Measurable impact. 10 of 56 threads · every case linked to its thread.

Data stops after the 4G to 2G change — Narrowed
Finding: control plane OK; loss at the MS–PCU boundary after PDP modify.
Next: handset-vs-PCU isolation test.
telecomhall.net/t/37073

Roaming subscriber receives no SMS or OTP — Advised
Finding: fault sits on the home-network side.
Check: ODB · stale VLR/MME registration · SoR.
telecomhall.net/t/36930

Which address carries S1-U? — Ongoing
Finding: endpoint by elimination, reservedBy → CUP_Inner.
Open: bind to the live S1-U F-TEID, UDP 2152.
telecomhall.net/t/37013

Before tuning a 5G SA handover failure — Corrected
Step 0: path, vendor + release, counter definition.
Top-rated thread · challenge accepted in public.
telecomhall.net/t/37018

EN-DC secondary-cell candidacy — Corrected
Cause: addSCellCandThresh read as an RSRP index.
Fix: real range is 1 to 100 per cent — corrected in public.
telecomhall.net/t/36886

S1 cause 29: interaction with other procedure — Ongoing
Finding: one sector, non-load factor at 54% DL PRB.
Open: RX branch imbalance on the RRU.
telecomhall.net/t/15161

Handover fails, re-establishment succeeds — Resolved
Cause: only a UTRA measObject; L800 never measured.
Fix: L800 EUTRA measObject + A3/A5 report.
telecomhall.net/t/36955

SRVCC B2 thresholds that did not add up — Corrected
Cause: 0.1 factor on hysteresisB2 — 1 dB, not 5 dB.
Next: UE trace before any further tuning.
telecomhall.net/t/36843

Shortage of NR TDD 5 MHz channels (CXX4012322) — Answered
Cause: licence capacity gate, not a fault — config exceeds key by 5 MHz.
Fix: remove 5 MHz or add capacity.
telecomhall.net/t/36820

Huawei BBU for an Ericsson engineer — Answered
Mapped moshell verbs to MML; read the DSP BRD on a BBU5900.
Result: TX on UMPT slot 7, CPRI on UBBP slots 1–2.
telecomhall.net/t/36924

Open source and community — Active
Open5GS issues #4690 · #4672 · #4699, S13/EIR proposal.
Osmocom #7048, SIGTRAN security defect.
github.com/open5gs · osmocom.org/issues/7048

Open5GS Discussions — Active
AKMA, SCP access control, steering of roaming, PCRF, CVE plan — support and community answers.

eUICC and JavaCard — Answered
GlobalPlatform domains, SCP02/SCP03 applet loading, SGP.32 SAIP, ISD-P isolation.

Lab and tooling built

- **5G SA reference lab** — AMF...SCP + gNB/UE · 6/6 baseline · ~12 s recovery
- **RF drive-test toolchain** — 3GPP validity ranges · PCI conflicts · 27 tests
- **NE access and evidence tool** — ZTE SSH/telnet · read-only gate · 21/21 tests
- **Signalling log triage** — Cause-value decode · 17 protocol families
- **AKMA, 3GPP Rel-17** — TS 33.535 vectors · ~3,557 tx/s · co-authored

Advisory

- **Connecting Kigali** — National 5G governance framework · MINICT, MININFRA
- **Current-state verification matrix** — 10 baseline confirmations · 5 institutions
- **Tower-estate energy baseline** — 1,467 shared sites · emissions and efficiency
- **Climate-finance briefs** — Lender, DFI and Rwanda Green Fund proposals

Publications and products

- **AI for Telecom Core Operations** — Five-volume handbook · 184 pages
- **Core Network Incident Playbook** — 18 playbooks · fix paths and verification gates
- **Field diagnostic casebook** — 22 pages · 14 full cases, linked to source
- **Felicia AI engine** — Evidence-locked RCA, MOP and evidence packs

Available on request

- ✓ MTN Rwanda CS core proposal · 2024
- ✓ SIGTRAN routing MOP, ABI Chennai/Mumbai · 2024
- ✓ VLR dual-active MOP, GAC/REM
- ✓ Extraction pipeline, first version · 2023 – 2025
- ✓ Performance report · 2024 – 2025
- ✓ Airtel Rwanda offer letter · 2025
- ✓ Full casebook (PDF) · master CV

REAL NETWORK PROBLEMS.
MEASURABLE RESULTS.

FELICIA
Your AI partner for deeper analysis, better decisions, bigger impact.

Prepared by Frutos
K. Frutos Kamanzi

Expertise + Experience + AI = Impact